

Mihir Patil

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Education

SVKM's DJSCE, B.Tech in Information Technology

Sept 2023 – May 2027

• **CGPA:** 8.725/10 | **honors:** DevOps

Technical Skills

Programming: C/C++, Python, Java, JavaScript, HTML5, CSS3, SQL

Frameworks: ReactJS, Express.js, Node.js, Flutter, TailwindCSS, Bootstrap, Next.js, Tauri

Tools: Git, GitHub, Figma, redis, MongoDB, Firebase, Docker, AWS, Jira, Supabase, langchain, langGraph

Projects

ASR-Based English Pronunciation Assistant (CDAC Project) (GitHub link)

Oct 2025 – May 26

Tech: PyTorch, Transformers, Deep Learning

- Developed a custom ASR model under CDAC inspired by Wav2Vec2, optimized for Indian English to improve pronunciation and speaking fluency.
- Designed and implemented a mispronunciation detection (MDD) module providing real-time, phoneme-level feedback.
- Enhanced speech clarity assessment by integrating transformer-based acoustic modeling and fine-tuned phoneme alignment pipelines.

ReflectOS - Personal AI Agent (GitHub link)

Nov 2025 – Jan 2026

Tech: Next.js, Python, OpenCV, YOLO, LangGraph

- Built a privacy-first AI smart mirror OS with a 42-node agentic architecture and hybrid memory system (Postgres, Redis, Vector DB).
- Enabled multi-modal interaction (voice, gestures, vision) with local-first processing for low-latency, secure automation.
- Designed a scalable LangGraph orchestration layer; reduced inference latency by 40% through optimized routing and LLM fallback strategies.

FedVision (GitHub link)

Aug 2025 – Jan 2026

Tech: Golang, Python, Mobilenet V2, Linux

- Designed and implemented a federated learning framework for computer vision across distributed edge computing environments.
- Optimized CNN training pipelines to efficiently utilize low-resource hardware, reducing overall training time by 50%.
- Achieved 90% model accuracy while enabling decentralized training across 2 edge devices.

KA-CHOW | AI-Driven SDLC Automation Platform (GitHub link)

Feb 2026

Tech: Next.js, Node.js, Neo4j, PostgreSQL, LangChain, CI/CD

- Architected a decoupled microservice platform using Next.js and Node.js to automate SDLC tasks, directly reducing manual toil and technical troubleshooting.
- Engineered a hybrid GraphRAG pipeline (Neo4j/PostgreSQL) for codebase intelligence, demonstrating advanced knowledge in AI.
- Automated CI/CD workflows and PR reviews via GitHub Webhooks to enforce secure, high-quality code and architectural stability across repositories.

Experience

Web Development Intern, Proxenix – New Delhi (Remote)

June 2025 – Aug 2025

- Selected for web development internship focused on modern frontend/backend technologies including React and Node.js .

Awards & Achievements

- **Winner (1st Prize)** at GDHS25; **1st Runner-up** at VEGA and a national hackathon; **2nd Runner-up** at Tech Expo.
- Competed in 15+ **hackathons** across AI and FinTech; secured **top-5 podium finishes** in major competitions.

Extracurricular Activities

- **DJCSI Committee Member (2023–24):** Publicity Co-Committee Member for Computer Society of DJ Sanghvi College
- **CodeShastra XI Organizing Team:** Part of organizing committee for Mumbai's one of the biggest hackathon